

Deploy next-gen space encryption hardware with fivefold performance boost for defense sovereign data protection, enabling secure satellite communications against quantum threats.

Defense × Data sovereignty and protection × Post-quantum cryptography · OS152 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
80 is the world moving	55 can we play, can we win	MEDIUM 0 named in the evidence	€103m observed evidence · per year	LATER research and standards activi...	L0 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 0.0 published reference(s) in Defense, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity deploys next-generation space encryption hardware with a fivefold performance boost to protect sovereign defense data against quantum threats. It targets defense organizations needing to secure satellite communications and data sovereignty now, as quantum computing advances and post-quantum cryptography standards emerge.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Observed: tendered contract value, annualised	€103m €103m – €196m	€103m €103m – €196m	€1.5m €1.5m – €2.9m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

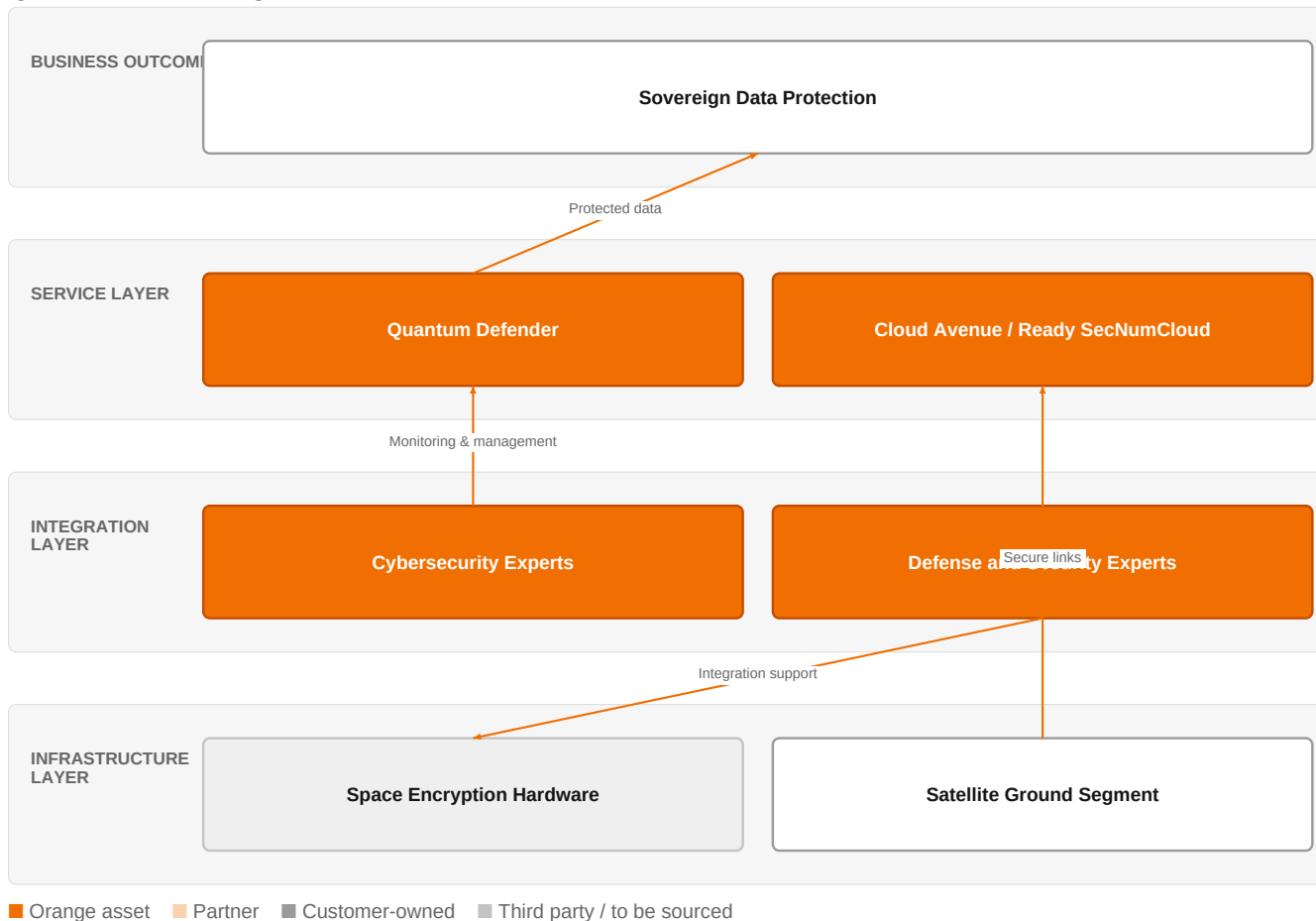
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€103m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-06
Assumed obtainable share of the serviceable market	1.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 19 matching notices, matched on use case via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 1 of 1 declared geographies are outside the European reference data (US); the estimate covers the European footprint only.

The solution, in one picture

Quantum-Safe Sovereign Satellite Communications



This diagram shows how Orange's quantum-safe offers and sovereign cloud integrate with third-party space encryption hardware to protect defense data.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Quantum computing advances threaten classical public-key cryptography, with algorithms like Shor's able to break RSA and elliptic-curve schemes, exposing data to harvest-now-decrypt-later attacks. NIST has standardized post-quantum cryptographic algorithms, prompting urgent migration across enterprise and defense systems. Space encryption hardware has received a fivefold performance boost, enabling practical quantum-resistant protection for satellite communications.

Sources: SIG-40B64E273856 International Journal 2026-07-30; SIG-B8561992126B openalex.org 2026-07-30; SIG-BFFA5CE48BF5 Shanlax International 2026-02-27; SIG-13AAE2CD702E MDPI AG 2026-01-19; SIG-0C2CE2735101 openalex.org 2025-10-13; SIG-7ECDD0A30D62 techtimes.com 2026-08-13

Who buys, and why now

The buyer is a defense procurement officer or CISO in a defense ministry or agency responsible for sovereign data protection. The pain is operational: current encryption on satellite links is vulnerable to quantum attacks, and adversaries may already be recording encrypted traffic for future decryption. The trigger is the NIST standardization of PQC algorithms and the availability of faster hardware, forcing a migration to quantum-resistant solutions.

Why it is hot now — every claim bound to a dated source

- Space encryption hardware receives a fivefold boost, enhancing data protection capabilities. [SIG-7ECDD0A30D62]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a quantum-safe communication solution using its Quantum Defender / quantum-safe offers, delivered on Cloud Avenue / Ready SecNumCloud for sovereign hosting. The engagement starts with a cryptographic asset discovery and inventory assessment (leveraging Orange Group researchers and cybersecurity experts), followed by a migration plan, then deployment of quantum-safe encryption on satellite ground segments and network links, with ongoing monitoring and management by Orange cybersecurity experts.

Why Orange can win

Orange brings certified sovereign infrastructure (SecNumCloud, ISO 27001) and deep defense expertise, which are critical for defense clients. The Quantum Defender / quantum-safe offers provide a ready-made solution, and Orange Group researchers contribute to migration strategy. However, the specific space encryption hardware is not an Orange asset, so Orange would need to partner or integrate third-party hardware, which could be a gap versus competitors with in-house hardware.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Quantum Defender / quantum-safe offers	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Cloud Avenue / Ready SecNumCloud	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Cybersecurity experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 6.3; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include Atos / Eviden, IBM, Thales, and Airbus Protect, all active in defense and selling post-quantum cryptography. Thales and Atos / Eviden have strong defense relationships and cloud/cybersecurity capabilities. IBM is a major player in quantum-safe solutions. Airbus Protect brings aerospace domain expertise. Entrust and ID Quantique are security specialists with PQC offerings. Dassault Systemes / 3DS Outscale offers sovereign cloud, and Dedrone by Axon is defense-focused. The customer will compare Orange's sovereign cloud and integration capabilities against these specialists.

Sources: SIG-7ECDD0A30D62 techtimes.com 2026-08-13

Competitor	Category	Position here	Basis	Evidence
Atos / Eviden	Global systems integrator	sells Post-quantum cryptography; active in Defense; competes in Cloud, Cybersecurity	structural	—
IBM	Global systems integrator	sells Post-quantum cryptography; competes in Cloud, Cybersecurity	structural	—
Airbus Protect	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cybersecurity	structural	—
Entrust	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—
ID Quantique	Security specialist	sells Post-quantum cryptography; competes in Cybersecurity	structural	—

Thales	Security specialist	sells Post-quantum cryptography; active in Defense; competes in Cloud, Cybersecurity	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Defense; competes in Cloud	structural	—
Dedrone by Axon	Security specialist	active in Defense; competes in Cybersecurity	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What satellite communication systems do you operate, and what encryption standards are currently in use?
- 2 Have you conducted a cryptographic asset inventory across your defense platforms?
- 3 What is your current timeline for post-quantum cryptography migration, and what triggers would accelerate it?
- 4 Do you have a requirement for sovereign or certified cloud infrastructure for your data protection solutions?
- 5 How are you currently addressing the harvest-now-decrypt-later threat for your sensitive communications?
- 6 What performance constraints exist on your satellite hardware that would affect the deployment of new encryption?

Objections you will meet

They will say	You can answer
We already have a post-quantum strategy with another vendor.	That's a valid approach. However, our sovereign cloud and certified infrastructure, combined with our defense expertise, can complement your existing strategy and ensure compliance with data sovereignty requirements.
Your solution lacks the specific space encryption hardware that competitors offer.	You're right that we don't manufacture that hardware. But we can integrate best-of-breed hardware with our quantum-safe offers and sovereign cloud, providing a complete solution with the assurance of our certifications.
Quantum computers are not a near-term threat, so why invest now?	The threat is real today through harvest-now-decrypt-later attacks. NIST has standardized PQC, and early migration reduces risk and cost. Our approach allows you to phase in quantum-safe measures without disrupting operations.

Risks and unknowns

The main risk is that Orange lacks the specific space encryption hardware, which could require partnering and add complexity. The migration of embedded defense systems is challenging, as noted in the evidence. The timeline for cryptographically relevant quantum computers is uncertain, which may affect urgency. Also, the performance of PQC on existing satellite hardware is not fully known, and the fivefold boost is from a competitor's announcement, not Orange's.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a defense satellite communications program that differentiates Orange by combining the Quantum Defender / quantum-safe offers with the SecNumCloud certification and ISO 27001, and include the Cybersecurity and Defense and security experts as key personnel.
Sales	In a conversation with a defense agency's chief information security officer, say: 'Given the quantum threat to satellite communications, we can help you protect sovereign data with our quantum-safe offers, which build on our SecNumCloud-certified cloud.'
Strategist / Innovator	Deep-dive the Quantum Defender / quantum-safe offers and the Cloud Avenue / Ready SecNumCloud offer to assess their fit for post-quantum cryptography in defense satellite communications, and map the relevant Cybersecurity and Defense and security experts to support a prototype.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-548F82402582	2026-08-01	Cryptography	1	Architecting Quantum-Resilient Blockchains: A Systems Framework for Post-Quantum Security, Gove...
SIG-40B64E273856	2026-07-30	International Journal of ...	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-B8561992126B	2026-07-30	openalex.org	1	Post-Quantum Cryptography Migration for Enterprise Security
SIG-BFFA5CE48BF5	2026-02-27	Shanlax International Jou...	1	Migration Strategies from Traditional Cryptography to Post-Quantum Cryptography in Cloud Systems
SIG-13AAE2CD702E	2026-01-19	MDPI AG	1	Cryptographic Asset Discovery and Inventory for Embedded Systems: A Framework for Post-Quantum ...
SIG-53F3508BD517	2026-01-09	openalex.org	1	Post-Quantum Public Key Infrastructures: Hybrid Certificates, Cryptographic Combiners, and Migr...
SIG-CE5A95F7AF72	2026-01-06	openalex.org	1	PQC Planner: Post-Quantum Cryptographic Migration Planner
SIG-0065C52738A8	2025-10-29	openalex.org	1	QRoNS: Quantum Resilience over IPsec Tunnels for Network Slicing
SIG-0C2CE2735101	2025-10-13	openalex.org	1	Post-Quantum Cryptography Migration of a Physical 5G Testbed
SIG-2153E369E0CC	2025-09-15	Journal of Electrical and...	1	Securing Digital Communications: Quantum and Post-Quantum Cryptography
SIG-43901F3A86B3	2025-04-09	Wiley	1	Post-Quantum Cryptography: Evaluating and Standardizing Quantum-Resistant Algorithms
SIG-2E8BDC1715A1	2026-08-18	Kalkine Media	2	Quantum Secure Encryption Unveils QSim to Streamline Enterprise Post-Quantum Cryptography Migra...
SIG-7ECDD0A30D62	2026-08-13	techtimes.com	2	Space Encryption Hardware Gets Fivefold Boost : Northrop Grumman and Aeronix

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:26+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.